YUFU CHEN

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EDUCATION

University of California, Berkeley Expected Spring 2028

B.A. in Cognitive Science

TECHNICAL SKILLS

Languages: Java, Python, HTML; C++ (working knowledge), Swift (project experience)

AI & Agent Workflows: AI-Assisted Development, Prompt Engineering, Agent Workflow Design, MCP Tool Integration

Tools & Technologies: Git, ARKit, visionOS, Unity, Blender

SELECTED PROJECTS

Aura — Assistive Spatial Computing for Apple Vision Pro | Swift, visionOS, ARKit July 2026

- Led product design and iterative prototyping of an accessibility-focused visionOS app; validated core functionality on Apple Vision Pro hardware.
- Validated ARKit scene-geometry and raycasting-based forward obstacle detection, proximity estimates, and spoken directional alerts; identified excessive warnings through device testing.

Reliable Memory System for Personalized AI Glasses | Python, SQLite, Pytest September 2026 – Present

- Developing cross-session memory infrastructure with structured storage, provenance tracking, entity/temporal scope validation, and memory retrieval for personalized AI assistants.
- Prototyped core Python memory modules and tests; designed baseline evaluations for retrieval reliability, token efficiency, and response latency (MVP in progress).

ModCraft — AI-Powered Minecraft Mod Generation | Co-Founder Oct 2025 – Jul 2026

- Co-founded a five-person team building an AI-powered Minecraft mod-generation tool; later made it freely available to members of our gaming community.
- Co-designed natural-language-to-DSL task decomposition and Generator/Builder system prompts; tested generated outputs, analyzed failures, and refined code-generation and repair prompts.

AI Smart Glasses — Computer Vision Prototype | Python, OpenCV, BLIP, CLIP 2024

- Prototyped a glasses-mounted camera system connected through an external development board to a PC-based computer vision pipeline, integrating pretrained models for scene description and visual-semantic matching.

sEMG Gesture Interface — Hands-Free Video Control | 2-Person Team September 2026 – Present

- Co-developing a wearable interaction prototype spanning four-channel forearm sEMG acquisition, signal processing, gesture classification, and hands-free short-video controls.
- Co-designing a pipeline to map five gesture states to seven UI commands using LDA/SVM decoding and a command state machine with voting, confidence rejection, and single-trigger locking (MVP in progress).

AIGF — Customizable AI Companion | Swift, iOS, macOS August 2026

- Designed and prototyped a customizable AI companion for iOS and macOS, exploring personalized conversational personas, Swift-based interfaces, and voice interaction.

LEADERSHIP

Co-President — Extended Reality (XR) at Berkeley Mar 2026 – Present

- Co-organize XR industry workshops and outreach, including XREAL events and prospective Snap Spectacles collaborations; engage approximately 15–20 students per workshop.
- Support Stanford XR hackathon collaboration and Berkeley DeCal programming introducing Unity, Blender, and XR game development.